# Blessed G/Michael Catholic Secondary School

## Semester 1 Biology Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

**Name:** _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

**I. Choose the best answer among the given alternatives.**

____ **1.** Which organelle is considered the "powerhouse of the cell" responsible for ATP synthesis?

**A.** Golgi apparatus **B.** Ribosome

**C.** Mitochondrion **D.** Lysosome

____ **2.** The movement of water molecules from a region of lower solute concentration to higher solute concentration across a semi-permeable membrane is:

**A.** Active transport **B.** Osmosis

**C.** Facilitated diffusion **D.** Endocytosis

____ **3.** During which phase of mitosis do sister chromatids separate and move toward opposite poles?

**A.** Prophase **B.** Metaphase **C.** Anaphase **D.** Telophase

____ **4.** In which stage of the cell cycle is DNA duplicated?

**A.** G1 phase **B.** S phase **C.** G2 phase **D.** M phase

____ **5.** Which biological branch specifically studies heredity and variation in living organisms?

**A.** Ecology **B.** Genetics **C.** Cytology **D.** Physiology

____ **6.** Which structure is present in plant cells but absent in typical animal cells?

**A.** Cell wall and chloroplast **B.** Ribosome and nucleus

**C.** Plasma membrane and cytoplasm **D.** Mitochondria and vacuole

____ **7.** Which of the following microscopes uses beams of electrons rather than light to achieve magnifications up to 1,000,000x?

**A.** Compound light microscope **B.** Transmission electron microscope (TEM)

**C.** Dissecting stereomicroscope **D.** Fluorescence microscope

____ **8.** What happens to a red blood cell placed into a hypertonic salt solution?

**A.** It absorbs water and lyses (bursts) **B.** It loses water and crenates (shrinks)

**C.** It remains unchanged in shape **D.** It divides by mitosis

____ **9.** Which component of the cytoskeleton forms the mitotic spindle fibers during nuclear cell division?

**A.** Microfilaments (actin) **B.** Intermediate filaments

**C.** Microtubules (tubulin) **D.** Collagen fibers

____ **10.** What is the end product of normal mitotic cell division?

**A.** Four genetically distinct haploid gametes
**B.** Two genetically identical diploid daughter cells
**C.** Two haploid zygotes
**D.** One multinucleated clone